



Published in final edited form as:

Nutr Cancer. 2023 ; 75(2): 552–561. doi:10.1080/01635581.2022.2135745.

Effect modification of selenium supplementation by intake and serum concentrations of antioxidants on the development of metachronous colorectal adenoma

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Abstract

Background: Selenium (Se) is a trace element that has been investigated as a potential chemopreventive agent for colorectal cancer. Dietary intake of other antioxidant nutrients may modify the effect of Se.

Objective: We examined the association between intake and serum concentrations of retinol, β -carotene, β -cryptoxanthin, lycopene, lutein/zeaxanthin, and α - and γ -tocopherol and the development of metachronous colorectal adenoma, and if these nutrients modified the effect of Se.

Methods: We conducted a prospective study of 1874 participants from the Se Trial with data for antioxidant intake, as well as a subcohort of 508 participants with serum biomarker concentrations.

Results: Statistically significantly lower odds for the development of metachronous adenoma were observed for those participants in the highest tertile of intake for lutein/zeaxanthin compared to the lowest, with an OR (95% CI) of 0.72 (0.56-0.94). No effect modification for intake of any nutrient was observed. However, circulating concentrations of lycopene exhibited statistically significant effect modification of selenium supplementation ($p < 0.06$),

Conclusion: These findings show that intake and circulating concentrations of antioxidant nutrients were not consistently associated with reduced odds for the development of metachronous

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Author Contributions: Primary data analysis and manuscript preparation was done by ETJ, with support from KS, MT, and MH. ETJ, PL, HHSC, JM, and KB made significant contributions to study design, data analysis and interpretation, and manuscript preparation. CC and WC generously contributed to this project analyzing blood levels of antioxidants, and by assisting with data interpretation and manuscript preparation.

Disclosure Statement: The authors report there are no competing interests to declare.

lesions, although blood concentrations of lycopene may modify the effect of selenium supplementation.

Keywords

Epidemiology; Antioxidant; Cancer Prevention

Introduction

Use of dietary supplements in the United States is common, with data from the National Health and Nutrition Examination Survey (NHANES) demonstrating that approximately half of adults use one or more dietary supplements, most commonly a multivitamin (1, 2). Though some supplement users do so with the expectation of reducing cancer risk, to date there is little evidence of a benefit in an average-risk population(3), and some indication that dietary supplements may be harmful in relation to cancer risk(4). Selenium is a trace element that has been studied extensively in the context of chemoprevention, with equivocal findings from randomized, controlled trials(5–7). Presently, there is a dearth of information regarding whether there may be interactions between selenium supplementation and other antioxidants.

One exception is the SELECT trial, which was designed to investigate whether selenium and α -tocopherol together may have greater effects on cancer development than either agent alone (8). The findings of the SELECT trial suggested that the impact of the combination of these two nutrients may have differed from that of either alone; the Se intervention was associated with an increased risk of type 2 diabetes, though this was not statistically significant, while the α -tocopherol supplemented group experienced an increase in risk for prostate cancer of a similar magnitude(5). However, in the treatment group where participants received both α -tocopherol and selenium, no increase in risk for either T2D or prostate cancer was reported. In addition, our group conducted an ancillary study within the SELECT trial and found that neither supplementation with selenium nor α -tocopherol reduced the risk of developing colorectal adenomas(9). These findings suggest the need to further study a potential interaction between these two nutrients, as well as between selenium and other antioxidants.

Antioxidant nutrients such as retinol, β -carotene, β -cryptoxanthin, lycopene, lutein/zeaxanthin, and α - and γ -tocopherol have long been hypothesized to be among the active components of fruits and vegetables that may account for the reduced risk observed for cancer among those with higher intakes of these foods (10). However, evidence for the impact of single antioxidants on risk for the development of cancer has been equivocal, with the results of randomized controlled trials indicating few, if any, benefits (3, 11–19).

To date, the potential for effect modification of Se by antioxidants other than α -tocopherol has not been investigated within the context of a randomized controlled trial. The Selenium Trial was a phase III randomized trial designed to determine whether selenium supplementation prevented adenoma recurrence. Data for antioxidant intake were also collected as part of the original trial, thus providing us with a unique opportunity to study modification of Se by antioxidants in depth in the Selenium Trial population;

additionally, stored samples enabled us to measure circulating concentrations of these nutrients. Therefore, the objective of this study was to determine whether dietary intake of antioxidants including retinol, β -carotene, β -cryptoxanthin, lycopene, lutein/zeaxanthin, and α - and γ -tocopherol, as well as serum biomarker concentrations of these nutrients, was associated with the development of metachronous adenoma, or modified the effect of Se supplementation on this outcome.

Methods

Study Population

All participants in the present study were those who were enrolled in the Selenium Trial, with a total of 1874 participants having data for antioxidant intake and development of metachronous adenomas. In addition, we randomly selected an antioxidant biomarker subcohort (n=508) from the overall pool of participants to analyze serum for circulating concentrations of the following biomarkers: retinol, β -carotene, β -cryptoxanthin, lycopene, lutein/zeaxanthin, and α - and γ -tocopherol. The Selenium Trial has been described in detail elsewhere (6, 20); briefly, it was a randomized, double-blind, placebo-controlled trial that tested the hypothesis that 200 μ g/d of Se as selenized yeast would significantly reduce the risk of metachronous adenoma compared to placebo. Participants were recruited from endoscopy clinics in Arizona, Colorado, Texas, and New York, and were healthy adult men and women between the ages of 40 and 80 years. Potential study subjects were required to have undergone a colonoscopy with removal of one or more colorectal adenomas with a diameter of 3 mm or more 6 months or less before study entry. Those with familial syndromes such as Lynch syndrome or familial adenomatous polyposis, or who had uncontrolled hypertension or heart disease, uncontrolled diabetes, or renal insufficiency, were excluded(6). Study data and biospecimens were collected and stored at the University of Arizona Cancer Center (Tucson, AZ)(20). The University of Arizona Institutional Review Board (IRB) approved the study, and conduct of the trial was in accordance with requirements of the local IRB at each study site.

Exposure and Outcome Assessment

Data regarding dietary, sociodemographic, and medical history data were collected with self-administered questionnaires completed by all participants in the Selenium Trial upon study entry. The Arizona Food Frequency Questionnaire (AFFQ) was employed for dietary data ascertainment. This instrument is a 113-item, semi-quantitative, scannable questionnaire that was modified from the National Cancer Institute's Health Habits and History Questionnaire(21). Participants were asked to report their intake of various foods for the prior year(22). Response categories ranging from >3 times/day to rarely/never were employed for most items, although for frequently-consumed foods and beverages, the scale ranged from >6 times/day to rarely/never(22). Total intake of each nutrient was calculated by multiplying the frequency of each item's consumption by the nutrient composition of each food(22).

Fat-soluble antioxidants were measured using HPLC, as has previously been performed by our group(23). Serum was mixed with an equal volume of ethanol [containing

0.1% butylated hydroxytoluene (BHT) antioxidant] for protein precipitation. Samples were then vortexed and analytes were extracted into hexane, evaporated under nitrogen, and redissolved in mobile phase (solvent A) before injecting onto the HPLC. Retinols, carotenoids, and tocopherols were separated by an ultrasphere ODS column (4.6-250 mm; Beckman Instruments) and detected at wavelengths of 325, 452, and 300 nm. The solvent system consisted of 95% solvent A and 5% solvent B at a flow rate of 2.5 mL/min. Solvent A was acetonitrile (ACN)-tetrahydrofuran (THF; 85:15, v/v) with 250 ppm BHT and 0.05% trimethylamine (TEA), and B was 50 mmol/L ammonium acetate in methanol with 0.05% TEA. Quantification of analytes was conducted by the external standard methods. Extinction coefficients were employed to spectrophotometrically validate the final solution concentrations. Standard reference material was supplied by the National Institute of Standards and Technology [NIST, Gaithersburg, MD] and were used for assigning values to in-house control materials. For lutein and zeaxanthin, prior work has demonstrated the difficulty in separating these two nutrients using common laboratory assays(24); therefore, they are combined here for both the analysis of circulating concentrations and for dietary intake.

Finally, the outcome of this study was the development of metachronous adenomas, defined as those detected upon a follow-up colonoscopy at least six months after study entry, and were categorized as either advanced or non-advanced(6). Advanced adenomas were defined as those with a diameter of 10 mm or greater, or having with villous features or high-grade dysplasia.

Statistical Analyses

Descriptive data for baseline characteristics of the overall Selenium Trial and the subcohort with measured serum levels of antioxidants were calculated using frequencies and percentages for categorical variables, and with means and standard deviations for the continuous variables. The association between dietary intake, serum concentrations of antioxidants, and the development of metachronous adenomas was assessed with unconditional logistic regression modeling, with an α of 0.05 for statistical significance. Variables assessed for potential confounding in both models were age, body mass index (BMI), sex, race, ethnicity, education, smoking status, and dietary intake of energy, protein, carbohydrate, fat, and fiber. Variables that changed the point estimate by 10% or greater were included in the final multivariate logistic regression analyses(25). Heterogeneity of effect for selenium supplementation by intake and serum levels of antioxidants was assessed by employing an interaction term for tertile of Se status and the variable in question and evaluating with a likelihood-ratio test, with statistical significance set at $p < 0.10$. All analyses were conducted using STATA statistical software package [version 17.0], Stata Corporation, College Station, TX].

Results

Characteristics for the overall population of participants from the Selenium Trial ($n=1874$) and the antioxidant subcohort ($n=508$) are presented in Table 1. For the overall trial population, characteristics including age, sex, body mass index, race, smoking history, and

Discussion

The results of this prospective study demonstrated that within the context of a clinical trial of 200 mcg/d of selenium vs. placebo, higher intake of lutein/zeaxanthin was significantly associated with a reduced risk for the development of any metachronous colorectal adenoma; while serum concentrations of retinol were significantly related to an increased risk for advanced lesions. When considering whether intake or serum levels of antioxidants modified the effect of selenium on the development of metachronous adenomas, heterogeneity was observed for circulating concentrations of lycopene, whereby selenium supplementation was statistically significantly associated with reduced odds for adenoma only among those with the highest serum concentrations of lycopene.

Selenium has long been the focus of chemoprevention trials since the publication of the Nutritional Prevention of Cancer (NPC) trial in 1996 (26). This double-blind, placebo-controlled trial randomized participants to either 200 µg Se per day as brewers' yeast or a matched placebo(26). The intervention did not reduce the risk of the primary endpoint for the trial, non-melanoma skin cancer, but in secondary analyses, a statistically significant 58% reduction in colorectal cancer incidence was observed for those receiving Se compared to placebo(26). When considering the impact of Se supplementation on colorectal adenomas in the trial, there was a suggestive reduction in risk among those in the intervention group compared to placebo(27). More importantly, this same report added to the growing field of literature indicating that individual participant characteristics may influence the efficacy of Se as a chemopreventive agent, as the results showed that any reduction in risk was confined to participants who entered the trial with blood Se levels in the lowest tertile (<105.5 ng/ml)(28). These findings dovetailed with prior reports from the NPC trial showing that the protection against overall cancer incidence conferred by Se supplementation was observed only among men and those with lower baseline Se blood levels(28).

Adding to the knowledge related to whether nutrients other than selenium may impact its efficacy in preventing colorectal neoplasia was the Selenium and Vitamin E Cancer Prevention Trial (SELECT), a randomized, double-blind, placebo-controlled trial conducted among more than 32,000 men and employing a 2x2 factorial design to examine the effect of 200 µg/d of Se as *L*-selenomethionine with and without 400 IU/d of vitamin E as *all* *rac*- α -tocopherol acetate on prostate cancer incidence(5, 29). No significant effect of Se or α -tocopherol was observed for preventing prostate cancer, nor for the additional endpoints of total, colorectal, or lung cancer incidence(5). In addition, a follow-up study within SELECT showed no reduction in the odds of developing metachronous lesions with either selenium or α -tocopherol(9). However, as mentioned previously, the primary results of the SELECT trial indicated that the combination of selenium and α -tocopherol may have had different effects than either nutrient alone raising the question of whether other antioxidant nutrients may have similar modifying effects.

We therefore sought to address this question in the context of the Selenium Trial, the primary findings of which revealed no significant effect of Se on the development of metachronous adenoma(6). The results of the most recent trials of selenium as a chemopreventive agent, therefore, have added more weight to the controversy regarding

cancer prevention strategies based on single-micronutrient supplementation(4, 30). Contrary to results for single-nutrient interventions for colorectal chemoprevention, dietary intake of fruits and vegetables has consistently been shown in observational studies to be associated with a reduced risk for colorectal adenomas and cancer(31–33), which in turn has led to numerous studies of the effects of single antioxidant nutrients found in these foods. However, intervention trials with single nutrients as well as with dietary patterns have largely failed to demonstrate protection from colorectal neoplasia(4, 34, 35).

In this work, we selected antioxidant nutrients to assess whether they were independently associated with the development of adenomas, as well as to determine if intake or blood concentrations modified the effect of selenium supplementation. We found that intake of lutein/zeaxanthin was statistically significantly related to a reduced risk for metachronous adenomas. These findings are in concordance with those of Mannisto et al.(36), who conducted a large meta-analysis of prospective studies of antioxidants and colorectal cancer, and found that the strongest reduction in risk was observed for lutein/zeaxanthin, though after adjustment for confounding, this relationship was not statistically significant. Lutein/zeaxanthin is primarily known for its role as key component in eye health and in vision but has been studied in relation to cancer for its efficiency at scavenging free radicals(36). Some prior studies have shown that higher circulating concentrations of this nutrient may be associated with a lower risk for colorectal cancer(37–40), while others have not(41, 42). *It has been proposed that oxidative stress may be elevated in adenomatous colorectal tissue, hence, the potentially heightened ability of lutein/zeaxanthin to neutralize free radicals may be a possible mechanism of action(36). In addition, it has been suggested that the localized action of this nutrient in the aqueous phase of the cell membrane may fortify membrane integrity to further protect the colorectal mucosa from free-radical damage(36).* Taken together, it appears that any impact of lutein/zeaxanthin on the development of colorectal neoplasia is modest.

Next, the finding that serum concentrations of retinol are significantly and directly related to increased odds of advanced metachronous lesions must be considered, though the comparatively few number of participants must be taken into account. Prior work has shown that lower blood levels of retinol were detected among colorectal cancer patients than among controls(43), and a similar finding was reported for colorectal polyps(44), but other investigations have shown no association between retinol and colorectal neoplasia(45). The findings here must be interpreted with caution due to the small number of participants in the lowest tertile for serum retinol levels; further, the potential toxicity of retinol makes it a poor nutrient to consider in relation to chemoprevention.

With regard to effect modification of selenium supplementation, only circulating concentrations of lycopene were found to be associated with heterogeneity of effect. As with lutein/zeaxanthin, the literature for the association between lycopene and colorectal neoplasia is equivocal(36, 40, 46, 47). It is unclear why selenium supplementation appeared to be more effective among those with higher blood concentrations of lycopene. One small, randomized clinical trial of selenium and lycopene together for the prevention of prostate cancer showed no effect of supplementation with these two nutrients(48), but another indicated that the combination of lycopene, selenium, and omega-3 fatty acids

combined was more effective at reducing PSA levels than any nutrient alone(49). Whether a combination of antioxidant nutrients is more effective than any single nutrient remains to be clarified, but the results of the present study suggest that for the majority of antioxidants and colorectal cancer, this is not the case.

The strengths of this study include that it was conducted in the context of a randomized clinical trial of selenium, as well as the large overall sample size, detailed dietary intake information, measured biomarkers of intake, and prospective design. However, limitations to the work must be acknowledged. First, when examining effect modification by dietary intake and biomarkers, some of the sample sizes were comparatively low. In addition, the participants included in the Selenium Trial were required to have had a prior colorectal adenoma to be eligible for the study. Therefore, it is possible that this study population was already more likely to develop colorectal neoplasia than the general population, leaving open the possibility that the reported associations may only be detected with metachronous adenoma. Finally, this study can provide no information regarding the role of antioxidants in later stages of carcinogenesis, as adenomas are precursor lesions to colorectal malignancies.

In conclusion, the results of this prospective analysis show a modest effect of higher dietary intake of lutein/zeaxanthin and lower risk for the development of metachronous adenomas, as well as effect modification of selenium supplementation by blood levels of lycopene. To our knowledge, these are the first findings to suggest that concentrations of other antioxidant nutrients may modify the effects of Se supplementation. Future work is needed to clarify whether lycopene and selenium together may have a greater impact on reducing the odds for metachronous adenomas than selenium alone. *It will also be important to measure the concentration of these nutrients within colorectal adenomatous tissue to ascertain whether intake and blood levels are directly associated with biological actions at the tissue level. Finally,, investigation of the association between these nutrients and biomarkers for metachronous colorectal adenoma, such as carcinoembryonic antigen and fibroblast growth factor-23 will help clarify a potential mechanism of action and targeted prevention efforts.*

Funding Information:

This work was supported by the National Institutes of Health under the following grants: National Cancer Institute under P30 CA023074, R01CA151708, P01 CA041108, and R01CA151708.

References

1. Kantor ED, Rehm CD, Du M, White E, Giovannucci EL. Trends in Dietary Supplement Use Among US Adults From 1999-2012. *JAMA Internal Medicine*. 2016;136(14):1464–74.
2. Gahche J, Bailey R, Burt V, Hughes J, Yetley E, Dwyer J, Picciano MF, McDowell M, Semplos C. Dietary supplement use among U.S. adults has increased since NHANES III (1988-1994). *NCHS Data Brief*. 2011(61):1–8. Epub 2011/05/20.
3. Gaziano JM, Sesso HD, Christen WG, Bubes V, Smith JP, MacFadyen J, Schvartz M, Manson JE, Glynn RJ, Buring JE. Multivitamins in the prevention of cancer in men: the Physicians' Health Study II randomized controlled trial. *JAMA*. 2012;308(18):1871–80. doi: 10.1001/jama.2012.14641. [PubMed: 23162860]

4. Martinez ME, Jacobs ET, Baron JA, Marshall JR, Byers T. Dietary supplements and cancer prevention: balancing potential benefits against proven harms. *Journal of the National Cancer Institute*. 2012;104(10):732–9. doi: 10.1093/jnci/djs195. [PubMed: 22534785]
5. Lippman SM, Klein EA, Goodman PJ, Lucia MS, Thompson IM, Ford LG, Parnes HL, Minasian LM, Gaziano JM, Hartline JA, Parsons JK, Bearden JD 3rd, Crawford ED, Goodman GE, Claudio J, Winquist E, Cook ED, Karp DD, Walther P, Lieber MM, Kristal AR, Darke AK, Arnold KB, Ganz PA, Santella RM, Albanes D, Taylor PR, Probstfield JL, Jagpal TJ, Crowley JJ, Meyskens FL Jr., Baker LH, Coltman CA Jr. Effect of selenium and vitamin E on risk of prostate cancer and other cancers: the Selenium and Vitamin E Cancer Prevention Trial (SELECT). *Jama*. 2009;301(1):39–51. [PubMed: 19066370]
6. Thompson PA, Ashbeck EL, Roe DJ, Fales L, Buckmeier J, Wang F, Bhattacharyya A, Hsu CH, Chow HH, Ahnen DJ, Boland CR, Heigh RI, Fay DE, Hamilton SR, Jacobs ET, Martinez ME, Alberts DS, Lance P. Selenium Supplementation for Prevention of Colorectal Adenomas and Risk of Associated Type 2 Diabetes. *Journal of the National Cancer Institute*. 2016;108(12). Epub 2016/08/18. doi: 10.1093/jnci/djw152.
7. Clark L, Combs G, al. e. Effects of selenium supplementation for cancer prevention for cancer prevention in patients with carcinoma of the skin: a randomized controlled trial. *JAMA*. 1996;276:1957–63. [PubMed: 8971064]
8. Lippman SM, Goodman PJ, Klein EA, Parnes HL, Thompson IM Jr., Kristal AR, Santella RM, Probstfield JL, Moynihan CM, Albanes D, Taylor PR, Minasian LM, Hoque A, Thomas SM, Crowley JJ, Gaziano JM, Stanford JL, Cook ED, Fleshner NE, Lieber MM, Walther PJ, Khuri FR, Karp DD, Schwartz GG, Ford LG, Coltman CA Jr. Designing the Selenium and Vitamin E Cancer Prevention Trial (SELECT). *Journal of the National Cancer Institute*. 2005;97(2):94–102. doi: 10.1093/jnci/dji009. [PubMed: 15657339]
9. Lance P, Alberts DS, Thompson PA, Fales L, Wang F, San Jose J, Jacobs ET, Goodman PJ, Darke AK, Yee M, Minasian L, Thompson IM, Roe DJ. Colorectal Adenomas in Participants of the SELECT Randomized Trial of Selenium and Vitamin E for Prostate Cancer Prevention. *Cancer prevention research*. 2017;10(1):45–54. Epub 2016/10/26. doi: 10.1158/1940-6207.CAPR-16-0104. [PubMed: 27777235]
10. Potter JD, Steinmetz K. Vegetables, fruit and phytoestrogens as preventive agents. *IARC Sci Publ*. 1996(139):61–90. Epub 1996/01/01.. [PubMed: 8923020]
11. Hercberg S, Galan P, Preziosi P, Bertrais S, Mennen L, Malvy D, Roussel AM, Favier A, Briancon S. The SU.VI.MAX Study: a randomized, placebo-controlled trial of the health effects of antioxidant vitamins and minerals. *Arch Intern Med*. 2004;164(21):2335–42. Epub 2004/11/24. doi: 10.1001/archinte.164.21.2335. [PubMed: 15557412]
12. Blot WJ, Li JY, Taylor PR, Guo W, Dawsey S, Wang GQ, Yang CS, Zheng SF, Gail M, Li GY, et al. Nutrition intervention trials in Linxian, China: supplementation with specific vitamin/mineral combinations, cancer incidence, and disease-specific mortality in the general population. *Journal of the National Cancer Institute*. 1993;85(18):1483–92. Epub 1993/09/15. doi: 10.1093/jnci/85.18.1483. [PubMed: 8360931]
13. Greenberg ER, Baron JA, Stukel TA, Stevens MM, Mandel JS, Spencer SK, Elias PM, Lowe N, Nierenberg DW, Bayrd G, et al. A clinical trial of beta carotene to prevent basal-cell and squamous-cell cancers of the skin. The Skin Cancer Prevention Study Group. *N Engl J Med*. 1990;323(12):789–95. Epub 1990/09/20. doi: 10.1056/NEJM199009203231204. [PubMed: 2202901]
14. Greenberg ER, Baron JA, Tosteson TD, Freeman DH Jr., Beck GJ, Bond JH, Colacchio TA, Collier JA, Frankl HD, Haile RW, et al. A clinical trial of antioxidant vitamins to prevent colorectal adenoma. Polyp Prevention Study Group. *N Engl J Med*. 1994;331(3):141–7. Epub 1994/07/21. doi: 10.1056/NEJM199407213310301.. [PubMed: 8008027]
15. Cole BF, Baron JA, Sandler RS, Haile RW, Ahnen DJ, Bresalier RS, McKeown-Eyssen G, Summers RW, Rothstein RI, Burke CA, Snover DC, Church TR, Allen JI, Robertson DJ, Beck GJ, Bond JH, Byers T, Mandel JS, Mott LA, Pearson LH, Barry EL, Rees JR, Marcon N, Saibil F, Ueland PM, Greenberg ER, Polyp Prevention Study G. Folic acid for the prevention of colorectal adenomas: a randomized clinical trial. *JAMA*. 2007;297(21):2351–9. Epub 2007/06/07. doi: 10.1001/jama.297.21.2351. [PubMed: 17551129]

16. Omenn GS, Goodman GE, Thornquist MD, Balmes J, Cullen MR, Glass A, Keogh JP, Meyskens FL, Valanis B, Williams JH, Barnhart S, Hammar S. Effects of a combination of beta carotene and vitamin A on lung cancer and cardiovascular disease. *N Engl J Med*. 1996;334(18):1150–5. Epub 1996/05/02. doi: 10.1056/NEJM199605023341802.. [PubMed: 8602180]
17. Alpha-Tocopherol BCCPSG. The effect of vitamin E and beta carotene on the incidence of lung cancer and other cancers in male smokers. *N Engl J Med*. 1994;330(15):1029–35. Epub 1994/04/14. doi: 10.1056/NEJM199404143301501. [PubMed: 8127329]
18. Hennekens CH, Buring JE, Manson JE, Stampfer M, Rosner B, Cook NR, Belanger C, LaMotte F, Gaziano JM, Ridker PM, Willett W, Peto R. Lack of effect of long-term supplementation with beta carotene on the incidence of malignant neoplasms and cardiovascular disease. *N Engl J Med*. 1996;334(18):1145–9. Epub 1996/05/02. doi: 10.1056/NEJM199605023341801. [PubMed: 8602179]
19. Taquet M, Dercon Q, Luciano S, Geddes JR, Husain M, Harrison PJ. Incidence, co-occurrence, and evolution of long-COVID features: A 6-month retrospective cohort study of 273,618 survivors of COVID-19. *PLoS Med*. 2021;18(9):e1003773. Epub 2021/09/29. doi: 10.1371/journal.pmed.1003773. following competing interests: SL is an employee of TriNetX Inc. The other authors report no interests to declare. [PubMed: 34582441]
20. Thompson P, Roe DJ, Fales L, Buckmeier J, Wang F, Hamilton SR, Bhattacharyya A, Green S, Hsu CH, Chow HH, Ahnen DJ, Boland CR, Heigh RI, Fay DE, Martinez ME, Jacobs E, Ashbeck EL, Alberts DS, Lance P. Design and baseline characteristics of participants in a phase III randomized trial of celecoxib and selenium for colorectal adenoma prevention. *Cancer prevention research*. 2012;5(12):1381–93. Epub 2012/10/13. doi: 10.1158/1940-6207.capr-12-0204. [PubMed: 23060037]
21. Block G, Hartman AM, Naughton D. A reduced dietary questionnaire: Development and validation. *Epidemiology*. 1990;1:58–64. [PubMed: 2081241]
22. Ritenbaugh C, Aickin M, Taren D, Teufel N, Graver E, Woolf K, Alberts DS. Use of a food frequency questionnaire to screen for dietary eligibility in a randomized cancer prevention phase III trial. *Cancer Epidemiol, Biomarkers & Prev*. 1997;6:347–54.
23. Hakim IA, Harris R, Garland L, Cordova CA, Mikhael DM, Sherry Chow HH. Gender difference in systemic oxidative stress and antioxidant capacity in current and former heavy smokers. *Cancer Epidemiol Biomarkers Prev*. 2012;21(12):2193–200. doi: 10.1158/1055-9965.EPI-12-0820. [PubMed: 23033455]
24. Goldbohm RA, Brants HA, Hulshof KF, van den Brandt PA. The contribution of various foods to intake of vitamin A and carotenoids in The Netherlands. *Int J Vitam Nutr Res*. 1998;68(6):378–83. Epub 1998/12/19. [PubMed: 9857265]
25. Mickey RM, Greenland S. The impact of confounder selection criteria on effect estimation. *American journal of epidemiology*. 1989;129(1):125–37. [PubMed: 2910056]
26. Clark LC, Combs GF Jr., Turnbull BW, Slate EH, Chalker DK, Chow J, Davis LS, Glover RA, Graham GF, Gross EG, Krongrad A, Leshner JL Jr., Park HK, Sanders BB Jr., Smith CL, Taylor JR. Effects of selenium supplementation for cancer prevention in patients with carcinoma of the skin. A randomized controlled trial. Nutritional Prevention of Cancer Study Group. *JAMA*. 1996;276(24):1957–63. [PubMed: 8971064]
27. Reid ME, Duffield-Lillico AJ, Marshall JR. Selenium supplementation and colorectal adenomas: an extended analysis of the Nutritional Prevention of Cancer Trial (abstract). *Frontiers in Cancer Prevention*. 2003:98.
28. Duffield-Lillico AJ, Reid ME, Turnbull BW, Combs GF, Slate EH, Fischbach LA, Marshall JR, Clark LC. Baseline characteristics and the effect of selenium supplementation on cancer incidence in a randomized clinical trial: A summary report of the Nutrition Prevention of Cancer Trial. *Cancer Epidemiol Biomarkers Prev*. 2002;11:630–9. [PubMed: 12101110]
29. Marshall JR, Sakr W, Wood D, Berry D, Tangen C, Parker F, Thompson I, Lippman SM, Lieberman R, Alberts D, Jarrard D, Coltman C, Greenwald P, Minasian L, Crawford ED. Design and progress of a trial of selenium to prevent prostate cancer among men with high-grade prostatic intraepithelial neoplasia. *Cancer Epidemiol Biomarkers Prev*. 2006;15(8):1479–84. doi: 10.1158/1055-9965.EPI-05-0585. [PubMed: 16896036]

30. Mayne ST, Ferrucci LM, Cartmel B. Lessons learned from randomized clinical trials of micronutrient supplementation for cancer prevention. *Annual review of nutrition*. 2012;32:369–90. doi: 10.1146/annurev-nutr-071811-150659.
31. Thompson PA, Ashbeck EL, Roe DJ, Fales L, Buckmeier J, Wang F, Bhattacharyya A, Hsu CH, Chow SH, Ahnen DJ, Boland CR, Heigh RI, Fay DE, Hamilton SR, Jacobs ET, Martinez EM, Alberts DS, Lance P. Celecoxib for the Prevention of Colorectal Adenomas: Results of a Suspended Randomized Controlled Trial. *Journal of the National Cancer Institute*. 2016;108(12). Epub 2016/08/18. doi: 10.1093/jnci/djw151.
32. Ubago-Guisado E, Rodriguez-Barranco M, Ching-Lopez A, Petrova D, Molina-Montes E, Amiano P, Barricarte-Gurrea A, Chirlaque MD, Agudo A, Sanchez MJ. Evidence Update on the Relationship between Diet and the Most Common Cancers from the European Prospective Investigation into Cancer and Nutrition (EPIC) Study: A Systematic Review. *Nutrients*. 2021;13(10). Epub 2021/10/24. doi: 10.3390/nu13103582.
33. Ben Q, Zhong J, Liu J, Wang L, Sun Y, Yv L, Yuan Y. Association Between Consumption of Fruits and Vegetables and Risk of Colorectal Adenoma: A PRISMA-Compliant Meta-Analysis of Observational Studies. *Medicine (Baltimore)*. 2015;94(42):e1599. Epub 2015/10/27. doi: 10.1097/MD.0000000000001599. [PubMed: 26496264]
34. Chapelle N, Martel M, Toes-Zoutendijk E, Barkun AN, Bardou M. Recent advances in clinical practice: colorectal cancer chemoprevention in the average-risk population. *Gut*. 2020;69(12):2244–55. Epub 2020/09/30. doi: 10.1136/gutjnl-2020-320990. [PubMed: 32989022]
35. Schatzkin A, Lanza E, Corle D, Lance P, Iber F, Caan B, Shike M, Weissfeld J, Burt R, Cooper MR, Kikendall JW, Cahill J. Lack of effect of a low-fat, high-fiber diet on the recurrence of colorectal adenomas. Polyp Prevention Trial Study Group. *N Engl J Med*. 2000;342(16):1149–55. Epub 2000/04/20. doi: 10.1056/NEJM200004203421601. [PubMed: 10770979]
36. Mannisto S, Yaun SS, Hunter DJ, Spiegelman D, Adami HO, Albanes D, van den Brandt PA, Buring JE, Cerhan JR, Colditz GA, Freudenheim JL, Fuchs CS, Giovannucci E, Goldbohm RA, Harnack L, Leitzmann M, McCullough ML, Miller AB, Rohan TE, Schatzkin A, Virtamo J, Willett WC, Wolk A, Zhang SM, Smith-Warner SA. Dietary carotenoids and risk of colorectal cancer in a pooled analysis of 11 cohort studies. *American journal of epidemiology*. 2007;165(3):246–55. Epub 2006/12/13. doi: 10.1093/aje/kwk009. [PubMed: 17158857]
37. Okuyama Y, Ozasa K, Oki K, Nishino H, Fujimoto S, Watanabe Y. Inverse associations between serum concentrations of zeaxanthin and other carotenoids and colorectal neoplasm in Japanese. *Int J Clin Oncol*. 2014;19(1):87–97. Epub 2013/02/06. doi: 10.1007/s10147-013-0520-2. [PubMed: 23380957]
38. Leung EY, Crozier JE, Talwar D, O'Reilly DS, McKee RF, Horgan PG, McMillan DC. Vitamin antioxidants, lipid peroxidation, tumour stage, the systemic inflammatory response and survival in patients with colorectal cancer. *International journal of cancer Journal international du cancer*. 2008;123(10):2460–4. Epub 2008/08/30. doi: 10.1002/ijc.23811. [PubMed: 18729200]
39. Nkondjock A, Ghadirian P. Dietary carotenoids and risk of colon cancer: case-control study. *International journal of cancer Journal international du cancer*. 2004;110(1):110–6. Epub 2004/04/01. doi: 10.1002/ijc.20066. [PubMed: 15054875]
40. Jung S, Wu K, Giovannucci E, Spiegelman D, Willett WC, Smith-Warner SA. Carotenoid intake and risk of colorectal adenomas in a cohort of male health professionals. *Cancer Causes Control*. 2013;24(4):705–17. Epub 2013/02/02. doi: 10.1007/s10552-013-0151-y. [PubMed: 23371557]
41. Kabat GC, Kim MY, Sarto GE, Shikany JM, Rohan TE. Repeated measurements of serum carotenoid, retinol and tocopherol levels in relation to colorectal cancer risk in the Women's Health Initiative. *Eur J Clin Nutr*. 2012;66(5):549–54. Epub 2011/12/15. doi: 10.1038/ejcn.2011.207. [PubMed: 22166899]
42. Terry P, Jain M, Miller AB, Howe GR, Rohan TE. Dietary carotenoid intake and colorectal cancer risk. *Nutr Cancer*. 2002;42(2):167–72. Epub 2002/11/06. doi: 10.1207/S15327914NC422_3. [PubMed: 12416255]
43. Starczak M, Zarakowska E, Modrzejewska M, Dziaman T, Szpila A, Linowiecka K, Guz J, Szpotan J, Gawronski M, Labejszo A, Liebert A, Banaszkiewicz Z, Klopocka M, Foksinski M, Gackowski D, Olinski R. In vivo evidence of ascorbate involvement in the generation of epigenetic DNA modifications in leukocytes from patients with colorectal carcinoma, benign

- adenoma and inflammatory bowel disease. *J Transl Med.* 2018;16(1):204. Epub 2018/07/22. doi: 10.1186/s12967-018-1581-9. [PubMed: 30029654]
44. Rumi G Jr., Szabo I, Vincze A, Matus Z, Toth G, Rumi G, Mozsik G. Decrease in serum levels of vitamin A and zeaxanthin in patients with colorectal polyp. *Eur J Gastroenterol Hepatol.* 1999;11(3):305–8. Epub 1999/05/20. doi: 10.1097/00042737-199903000-00015. [PubMed: 10333204]
45. Jiang J, Suzuki S, Xiang J, Kuriki K, Hosono A, Arakawa K, Wang J, Nagaya T, Kojima M, Katsuda N, Tokudome S. Plasma carotenoid, alpha-tocopherol and retinol concentrations and risk of colorectal adenomas: A case-control study in Japan. *Cancer Lett.* 2005;226(2):133–41. Epub 2005/05/12. doi: 10.1016/j.canlet.2005.03.040. [PubMed: 15885891]
46. Erhardt JG, Meisner C, Bode JC, Bode C. Lycopene, beta-carotene, and colorectal adenomas. *Am J Clin Nutr.* 2003;78(6):1219–24. Epub 2003/12/12. doi: 10.1093/ajcn/78.6.1219. [PubMed: 14668286]
47. Huang C, Wang Y, Li X, Ren L, Zhao J, Hu Y, Zhang L, Fan G, Xu J, Gu X, Cheng Z, Yu T, Xia J, Wei Y, Wu W, Xie X, Yin W, Li H, Liu M, Xiao Y, Gao H, Guo L, Xie J, Wang G, Jiang R, Gao Z, Jin Q, Wang J, Cao B. Clinical features of patients infected with 2019 novel coronavirus in Wuhan, China. *Lancet.* 2020;395(10223):497–506. Epub 2020/01/28. doi: 10.1016/S0140-6736(20)30183-5. [PubMed: 31986264]
48. Morgia G, Voce S, Palmieri F, Gentile M, Iapicca G, Giannantoni A, Blefari F, Carini M, Vespasiani G, Santelli G, Arnone S, Pareo RM, Russo GI. Association between selenium and lycopene supplementation and incidence of prostate cancer: Results from the post-hoc analysis of the procomb trial. *Phytomedicine.* 2017;34:1–5. Epub 2017/09/14. doi: 10.1016/j.phymed.2017.06.008. [PubMed: 28899491]
49. Paur I, Lilleby W, Bohn SK, Hulander E, Klein W, Vlatkovic L, Axcrona K, Bolstad N, Bjoro T, Laake P, Tasken KA, Svindland A, Eri LM, Brennhovd B, Carlsen MH, Fossa SD, Smeland SS, Karlsen AS, Blomhoff R. Tomato-based randomized controlled trial in prostate cancer patients: Effect on PSA. *Clin Nutr.* 2017;36(3):672–9. Epub 2016/07/14. doi: 10.1016/j.clnu.2016.06.014. [PubMed: 27406859]

Table 1.

Baseline characteristics of participants in the Selenium Trial and antioxidant sub-cohort, by treatment group.

Characteristic	Overall Se Trial		Antioxidant sub-cohort	
	Placebo Group n=914	Se Group N=910	Placebo Group N=256	Se Group N=252
Age (years, mean $\pm$ sd)	62.6 $\pm$ 8.9	63.2 $\pm$ 9.0	63.1 $\pm$ 9.1	64.1 $\pm$ 8.8
Male (n, %)	603 (51.2)	576 (49.0)	171 (51.0)	164 (49.0)
BMI in kg/m ² (n, %) ¹				
<25	196 (48.2)	189 (51.7)	47 (51.7)	44 (48.4)
25 to <30	406 (51.5)	383 (48.5)	111 (49.3)	114 (50.7)
$\geq$ 30	332 (50.0)	338 (50.0)	98 (51.0)	94 (49.0)
Race (n, %) ¹				
White	851 (49.8)	857 (50.1)	237 (50.0)	241 (50.0)
Black	26 (47.3)	29 (52.7)	5 (45.6)	6 (54.6)
Ever smoked (yes, n, %) ¹	497 (48.2)	535 (51.8)	131 (48.5)	139 (51.5)
Dietary intake				
Energy (kcal/d)	1793.2 $\pm$ 806.1	1824.9 $\pm$ 935.6	1774.7 $\pm$ 828.7	1794.3 $\pm$ 895.5
Protein (g/d)	72.8 $\pm$ 32.8	74.4 $\pm$ 40.6	71.7 $\pm$ 33.0	73.5 $\pm$ 35.5
Carbohydrate (g/d)	239.6 $\pm$ 118.4	241.3 $\pm$ 130.5	241.0 $\pm$ 119.5	234.7 $\pm$ 130.8
Total fat (g/d)	59.4 $\pm$ 31.1	60.8 $\pm$ 36.0	58.4 $\pm$ 32.4	59.4 $\pm$ 34.4
Total fiber (g/d)	20.0 $\pm$ 10.8	20.5 $\pm$ 11.9	20.2 $\pm$ 10.3	20.1 $\pm$ 11.0

Table 2.

Association between tertiles of dietary intake and total intake of antioxidant nutrients and colorectal adenoma recurrence in the Selenium Trial.

Dietary Intake tertile Mean intake/d $\pm$ sd	Any recurrence		Advanced recurrence	
	Recurrence n=639	aOR (95% CI) ¹	Recurrence n=137	aOR (95% CI) ¹
Retinol (mcg/d)				
198.7 $\pm$ 68.8	213	1.00	49	1.00
408.8 $\pm$ 60.3	212	0.88 (0.64-1.21)	41	0.72 (0.46-1.13)
823.8 $\pm$ 592.8	214	1.01 (0.74-1.39)	47	0.79 (0.51-1.23)
Lutein/zeaxanthin (mcg/d)				
875.6 $\pm$ 295.9	228	1.00	48	1.00
1901.5 $\pm$ 352.2	228	0.99 (0.77-1.28)	50	0.99 (0.65-1.52)
4452.6 $\pm$ 2071.2	183	0.72 (0.56-0.94)	39	0.84 (0.54-1.32)
β-cryptoxanthin (mcg/d)				
41.0 $\pm$ 60.5	233	1.00	49	1.00
117.3 $\pm$ 32.7	202	0.73 (0.54-1.00)	41	0.72 (0.46-1.13)
369.2 $\pm$ 177.0	204	0.78 (0.56-1.07)	47	0.83 (0.54-1.29)
γ-tocopherol (mg/d)				
1.62 $\pm$ 0.52	225	1.00	44	1.00
3.25 $\pm$ 0.49	215	0.87 (0.63-1.19)	46	0.89 (0.57-1.40)
6.69 $\pm$ 2.8	199	0.73 (0.52-1.01)	47	0.90 (0.57-1.42)
α-tocopherol (mg/d)				
3.7 $\pm$ 0.99	229	1.00	47	1.00
6.3 $\pm$ 0.80	208	0.70 (0.51-0.96)	46	0.85 (0.55-1.31)
12.0 $\pm$ 6.1	202	0.79 (0.58-1.08)	44	0.82 (0.53-1.28)
Lycopene (mcg/d)				
2072.4 $\pm$ 738.9	215	1.00	53	1.00
4158.4 $\pm$ 574.9	226	1.08 (0.79-1.48)	45	0.73 (0.48-1.13)
8367.1 $\pm$ 3552.2	198	0.83 (0.60-1.16)	39	0.65 (0.41-1.01)
α-carotene (mcg/d)				
150.2 $\pm$ 59.8	230	1.00	48	1.00
375.2 $\pm$ 87.9	207	0.82 (0.65-1.21)	47	0.94 (0.61-1.45)
1297.7 $\pm$ 1927.3	202	0.81 (0.59-1.11)	42	0.85 (0.55-1.33)
β-carotene (mcg/d)				
1156.1 $\pm$ 403.3	238	1.00	55	1.00
2486.2 $\pm$ 446.6	211	0.85 (0.63-1.16)	39	0.67 (0.43-1.05)
6097.9 $\pm$ 5313.8	190	0.65 (0.48-1.90)	43	0.76 (0.50-1.17)

¹ Models adjusted for age, sex, and history of colorectal polyps prior to study entry.

² Total intake includes dietary and supplemental intake.

Table 3.

Association between tertiles of plasma concentrations of antioxidant nutrients and colorectal adenoma recurrence in the Selenium Trial.

Biomarker concentration tertile Mean $\pm$ sd ($\mu\text{g/ml}$)	Any recurrence		Advanced recurrence	
	Recurrence n=238	OR (95% CI) ¹	Recurrence	OR (95% CI) ¹
Retinol				
0.49 $\pm$ 0.10	78	1.00	7	1.00
0.70 $\pm$ 0.05	96	1.38 (0.89-2.16)	21	2.97 (1.20-7.32)
1.00 $\pm$ 0.21	64	0.68 (0.44-1.06)	17	2.55 (1.02-6.34)
Lutein/zeaxanthin				
0.08 $\pm$ 0.01	81	1.00	11	1.00
0.17 $\pm$ 0.03	92	1.33 (0.86-2.07)	19	1.96 (0.88-4.37)
0.36 $\pm$ 0.17	64	0.67 (0.43-1.04)	15	1.41 (0.60-3.29)
β-cryptoxanthin				
0.02 $\pm$ 0.03	83	1.00	12	1.00
0.05 $\pm$ 0.01	84	1.10 (0.71-1.70)	14	1.14 (0.50-2.59)
0.14 $\pm$ 0.18	71	0.80 (0.53-1.29)	19	1.76 (0.81-3.81)
γ-tocopherol ($\mu\text{g/ml}$)				
0.47 $\pm$ 0.27	77	1.00	15	1.00
1.44 $\pm$ 0.33	86	1.31 (0.85-2.05)	16	1.19 (0.55-2.55)
3.62 $\pm$ 1.92	75	1.04 (0.67-1.63)	14	0.96 (0.43-2.14)
α-tocopherol ($\mu\text{g/ml}$)				
10.0 $\pm$ 2.02	85	1.00	17	1.00
15.8 $\pm$ 1.80	81	0.87 (0.56-1.35)	9	0.44 (0.18-1.06)
29.0 $\pm$ 19.8	72	0.65 (0.41-1.01)	19	1.02 (0.50-2.09)
Lycopene ($\mu\text{g/ml}$)				
0.16 $\pm$ 0.08	80	1.00	10	1.00
0.42 $\pm$ 0.09	79	1.00 (0.64-1.54)	16	1.78 (0.75-4.22)
1.04 $\pm$ 0.52	79	1.10 (0.71-1.72)	19	2.46 (1.06-5.68)
α-carotene ($\mu\text{g/ml}$)				
0.01 $\pm$ 0.007	79	1.00	14	1.00
0.05 $\pm$ 0.01	85	1.17 (0.76-1.82)	14	1.08 (0.49-2.37)
0.45 $\pm$ 0.78	74	0.90 (0.58-1.40)	17	1.35 (0.62-2.91)
β-carotene ($\mu\text{g/ml}$)				
0.08 $\pm$ 0.04	74	1.00	14	1.00
0.20 $\pm$ 0.04	81	1.24 (0.79-1.93)	11	0.88 (0.38-2.05)
0.64 $\pm$ 0.44	83	1.24 (0.79-1.93)	20	1.61 (0.76-3.43)
Retinyl palmitate ($\mu\text{g/ml}$)				
0.01 $\pm$ 0.004	77	1.00	16	1.00
0.02 $\pm$ 0.003	79	1.08 (0.67-1.63)	8	0.50 (0.21-1.23)
0.04 $\pm$ 0.01	82	1.13 (0.73-1.76)	21	1.38 (0.67-2.83)

¹Models adjusted for age, sex, and history of colorectal polyps prior to study entry.

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Odds ratios and 95% confidence intervals for the effect of selenium supplementation vs. placebo on overall adenoma on the development of metachronous adenoma, stratified by tertile of antioxidant intake.

Table 4.

OR (95% CI) for selenium supplementation vs. placebo and metachronous adenoma					
	Tertile 1	Tertile 2	Tertile 3	p-interaction	
Dietary Intake	Mean ± sd, (n metachronous/total)	Mean ± sd, (n metachronous/total)	Mean ± sd, (n metachronous/total)		
Retinol (mcg/d)	198.7 ± 68.8	408.8 ± 60.3	823.8 ± 592.8		
OR (95% CI)	1.40 (0.86-2.30)	0.75 (0.46-1.22)	0.69 (0.42-1.15)	0.57	
Lutein/zeaxanthin (mcg/d)	875.6 ± 295.9	1901.5 ± 352.2	4452.6 ± 2071.2		
OR (95% CI)	0.59 (0.37-0.95)	1.24 (0.76-2.01)	0.72 (0.43-1.20)	0.90	
β-cryptoxanthin (mcg/d)	41.0 ± 60.5	117.3 ± 32.7	369.2 ± 177.0		
OR (95% CI)	1.36 (0.82-2.24)	0.56 (0.35-0.88)	1.03 (0.61-1.76)	0.08	
γ-tocopherol (mg/d)	1.62 ± 0.52	3.25 ± 0.49	6.69 ± 2.8		
OR (95% CI)	1.07 (0.67-1.71)	0.82 (0.49-1.38)	0.65 (0.39-1.08)	0.69	
α-tocopherol (mg/d)	3.7 ± 0.99	6.3 ± 0.80	12.0 ± 6.1		
OR (95% CI)	1.14 (0.72-1.81)	0.47 (0.28-0.77)	1.43 (0.83-2.48)	0.80	
Lycopene (mcg/d)	2072.4 ± 738.9	4158.4 ± 574.9	8367.1 ± 3552.2		
OR (95% CI)	0.98 (0.62-1.57)	0.91 (0.59-1.41)	0.82 (0.52-1.31)	0.80	
α-carotene (mcg/d)	150.2 ± 59.8	375.2 ± 87.9	1297.7 ± 1927.3		
OR (95% CI)	1.04 (0.65-1.66)	0.97 (0.61-1.54)	0.70 (0.41-1.19)	0.48	
β-carotene (mcg/d)	1156.1 ± 403.3	2486.2 ± 446.6	6097.9 ± 5313.8		
OR (95% CI)	0.59 (0.37-0.95)	1.24 (0.76-2.01)	0.95 (0.56-1.61)	0.42	

Odds ratios and 95% confidence intervals for the effect of selenium supplementation vs. placebo on overall adenoma on the development of metachronous adenoma, stratified by tertile of serum biomarker concentrations.

Table 5.

OR (95% CI) for selenium supplementation vs. placebo, and metachronous adenoma				
	Tertile 1	Tertile 2	Tertile 3	p-interaction
Mean ± sd, (n metachronous/total)				
Mean ± sd, (n metachronous/total)				
Blood Biomarkers				
Retinol (µg/ml)	0.49 ± 0.10	0.70 ± 0.05	1.00 ± 0.21	
OR (95% CI)	1.61 (1.01-2.55)	0.47 (0.27-0.79)	0.92 (0.56-1.52)	0.30
Lutein/zeaxanthin (µg/ml)	0.08 ± 0.01	0.17 ± 0.03	0.36 ± 0.17	
OR (95% CI)	0.82 (0.48-1.40)	0.99 (0.61-1.60)	0.72 (0.43-1.20)	0.78
β-cryptoxanthin (µg/ml)	0.02 ± 0.03	0.05 ± 0.01	0.14 ± 0.18	
OR (95% CI)	1.36 (0.82-2.24)	0.56 (0.35-0.88)	1.03 (0.61-1.76)	0.44
γ-tocopherol (µg/ml)	0.47 ± 0.27	1.44 ± 0.33	3.62 ± 1.92	
OR (95% CI)	1.50 (0.89-2.54)	1.42 (0.88-2.29)	0.49 (0.30-0.82)	0.12
α-tocopherol (µg/ml)	10.0 ± 2.02	15.8 ± 1.80	29.0 ± 19.8	
OR (95% CI)	0.99 (0.61-1.63)	1.05 (0.63-1.75)	1.00 (0.95-1.06)	0.62
Lycopene (µg/ml)	0.16 ± 0.08	0.42 ± 0.09	1.04 ± 0.52	
OR (95% CI)	1.18 (0.75-1.86)	1.13 (0.69-1.85)	0.51 (0.30-0.88)	0.06
α-carotene (µg/ml)	0.01 ± 0.007	0.05 ± 0.01	0.45 ± 0.78	
OR (95% CI)	0.66 (0.40-1.08)	1.07 (0.66-1.75)	1.01 (0.61-1.68)	0.87
β-carotene (µg/ml)	0.08 ± 0.04	0.20 ± 0.04	0.64 ± 0.44	
OR (95% CI)	1.33 (0.83-2.14)	0.43 (0.26-0.71)	1.29 (0.78-2.14)	0.93